

Political Violence, Memory, and National Bias in Consumption: Evidence from Postwar Italy*

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Abstract

This paper examines the effects of wartime violence on consumer preferences for durable goods. We exploit cross-provincial variation in civilian casualties in Italy during 1943-1945, arising from Nazi occupation and the final phase of Fascist rule, to study the post-WWII purchase patterns of (imported) German vehicles. Using panel regressions with province and year fixed effects, we estimate how exposure to violence, interacted with local electoral shares of the Italian Communist Party, the institutional carrier of wartime memory, affects the composition of vehicle purchases. Our results show that higher exposure to wartime violence reduces the share of German passenger cars among foreign cars, with the effect concentrated on authentically German brands and attenuated when American-owned Ford and Opel vehicles manufactured in Germany are included. In contrast, we do not find any effect on commercial trucks, consistent with persistent preference shifts that shape household consumption but leave business demand unaffected.

Keywords: Political violence, Collective memory, WWII, Imports, Automobiles.

JEL Codes: D12, F14, N44.

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1 Introduction

On 8 September 1943, Italy signed an armistice with the Allied powers. The German army rapidly took control of the Center and North of the country, disarmed the Italian soldiers, and reinstalled Mussolini as the head of the Nazi-backed *Repubblica Sociale Italiana* (RSI, “Italian Social Republic”, also known as the Salò Republic).¹ Over the next twenty months, the German army and its Fascist collaborators carried out a “retaliatory” campaign of violence against civilians that historians have characterised as a civil war as much as an occupation (Pavone, 2014): 5,910 episodes of violence, totalling more than 25,000 civilian casualties. Some acute episodes, counting hundreds of victims, became Italian national symbols of Nazi occupation. Many others were smaller, local, and less visible outside the communities directly affected. Yet together they produced a geography of wartime atrocities and suffering that was highly uneven across Italian provinces and that remained part of local social memory long after the war had ended.

Can such historical violence shape economic decisions decades later? More specifically, can the memory of the massacres committed by a foreign country occupant affect the national origin of goods that households choose to buy in peacetime? These questions are central to understanding how history influences consumers’ choices. Standard accounts of postwar recovery emphasize reconstruction, income growth, and the restoration of trade. Once destroyed infrastructure is rebuilt and supply constraints are overcome, wartime shocks should gradually lose relevance for ordinary market behaviour. Yet many economic decisions involve goods whose perceived characteristics are often not exhausted by price and quality. Durable consumer goods, and cars in particular, are visible objects, associated with status, identity, and national reputation. If a foreign country is remembered not only as a trading partner but also as a former perpetrator, the national origin of its goods may carry a subjective cost.

This paper studies these issues in postwar Italy. We ask whether civilian casualties caused by Nazi-fascist violence during 1943–1945 affected subsequent purchases of German passenger cars. Our empirical setting combines three features that make

¹Terms in Italian will be identified by italics.

this question unusually tractable. First, the violence was geographically heterogeneous. Because the German occupation lasted far longer in the Center and North than in the South, provinces differed sharply in their exposure to civilian killings. Reprisals also followed the military geography of the front lines and of the armed partisan response, the *Resistenza* (Cooke, 1997; Pavone, 2014). Second, the postwar Italian automobile market provides detailed administrative information on consumer choices. Using annual publications of the *Automobile Club d'Italia* (ACI; the national agency in charge of the registration system), we compile province-level data on new vehicle registrations by type, country of origin, and brand for national election years from the 1950s to the early 1990s. Third, postwar Italy had a mass political organization with both the incentive and the capacity to keep wartime memory politically active: *Partito Comunista Italiano* (PCI; Italian Communist Party).

The mechanism we emphasize is not a direct mechanical link from violence to later consumption. Historical shocks do not automatically survive for decades. Memories can fade, be reinterpreted, become privatized within families, or be displaced by new economic and political concerns. Our argument is instead that the PCI acted as an institutional carrier of wartime memory. In provinces where the party was stronger, Nazi-fascist violence was more likely to be publicly commemorated, politically interpreted, and incorporated into a broader anti-fascist and anti-German discourse. Annual Liberation Day ceremonies, local commemorations of massacres, the party press, ANPI (National Association of Italian Partisans) events, and the dense network of local party sections all contributed to maintaining the salience of wartime suffering. In this sense, PCI vote share is more than a simple measure of individual Communist preferences and electoral success. It captures the strength of a local organizational infrastructure capable of sustaining a public discourse around the memory of Nazi-fascist violence.²

We test this argument by exploiting cross-provincial variation in civilian casualties and its interaction with PCI electoral strength across ten national elections spanning the years 1953-1992. Our benchmark specification relates the share of German passenger

²The mechanism does not require PCI voters to consciously boycott German goods, nor does it require anti-German sentiment to be confined to PCI voters. The relevant exposure is provincial: residents of high-PCI provinces were more likely to live in a local public environment in which wartime violence was repeatedly commemorated and politically framed.

cars among foreign cars in province c and year t to the interaction between province-level wartime casualties and the PCI vote share in that province and election year. Province fixed effects absorb all time-invariant local characteristics, including the direct effect of casualty exposure, persistent political culture, and long-run differences in income, urbanization, human capital or market access. Year fixed effects absorb common shocks to the Italian car market, including national business-cycle conditions, changes in trade policy, and shifts in the relative attractiveness of German cars. The coefficient of interest therefore asks whether, within a province, German car purchases are lower in years in which the local political carrier of wartime memory is stronger, and whether this inverse relationship is accentuated in provinces more exposed to 1943-1945 Nazi-fascist violence.

The main result is that political violence is associated with a persistent reduction in German car purchases, but only where the political memory mechanism is strong enough. This result is robust across several measures of violence, including casualties per capita, the number of casualties per event, terciles of massacre intensity, and the logarithm of total casualties. The magnitude is economically meaningful. At the average PCI vote share, a one-standard-deviation increase in casualties per event reduces the German share of foreign cars by 9.5% of its sample mean.

Several observations support the interpretation that the result reflects a national-origin preference rather than a generic demand shock. First, the effect is concentrated on German-owned brands. ACI records country of production, which implies that Ford and Opel cars manufactured in Germany appear as German imports. Yet Italian consumers would plausibly have associated these cars with American rather than German ownership. When we exclude Ford and Opel vehicles imported from Germany, the estimated effects become stronger and more systematic. This brand specificity is difficult to reconcile with explanations based only on supply conditions or the physical origin of the vehicle; it is consistent with a consumer sentiment channel directed at goods perceived as German. The bias we document is not the familiar “buy domestic” pattern. It is directed outward, against a specific foreign nation, rather than inward in favour of domestic production.

Second, we find no evidence that casualties reduced the overall propensity to buy foreign cars. The interaction term between casualties and PCI vote share does not lower the share of foreign cars in total registrations. The effect operates within the foreign segment of the market, changing its composition rather than shifting consumers away from imported goods altogether. This matters because it helps distinguish anti-German sentiment from generic economic nationalism or “buy Italian” behaviour. Consumers in high-casualty, high-PCI provinces did not simply avoid imports; they avoided German-branded passenger cars.

Third, the effect is absent for commercial trucks. We replicate the same specifications using the share of German trucks among foreign truck registrations and find no comparable pattern. This placebo is central to the interpretation. If the results were driven by supply disruptions, local procurement constraints, discriminatory access to dealerships, or broad economic differences across provinces, one would expect similar effects for commercial vehicles. Instead, the same historical exposure and the same political interaction that predict passenger-car choices do not predict truck purchases. Since trucks were purchased primarily by firms for productive use, while passenger cars were household durable goods, the contrast points toward household taste and publicly visible identity rather than business demand or supply-side frictions.

The paper contributes to four connected literatures. First, it speaks to work on the long-run consequences of war and political violence. Existing evidence shows that wartime destruction need not produce permanent material scarring when reconstruction forces are strong. Our results point to a different margin of persistence: even when markets recover and trade resumes, the memory of violence may continue to affect the composition of demand. Second, the paper contributes to the literature on historical persistence and cultural transmission by identifying an institutional channel through which a historical shock remains active. The relevant object is not persistence alone, but politically mediated persistence. Third, we add to research on endogenous preferences and home bias in trade by showing that national-origin preferences can be shaped by violent historical experience and sustained by local political institutions. Finally, the paper contributes to the economic literature on fascism and its legacies. Much

of that literature studies the origins of fascist support, the persistence of neo-fascist political cultures, or the effect of Fascist institutions on later outcomes. We study a complementary legacy: how Nazi-fascist violence, preserved in anti-fascist political memory, affected private consumption choices after the dictatorship had collapsed.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 provides the historical background on the German occupation, Nazi-fascist violence, and the geography of civilian casualties. Section 4 develops the mechanism through which the PCI kept wartime memory politically alive. Section 5 describes the data on casualties, vehicle registrations, electoral outcomes, and provincial controls. Section 6 presents the empirical strategy. Section 7 reports the main results, substitution patterns, robustness checks, and the commercial-truck placebo. Section 8 concludes.

2 Literature review

This paper sits within the growing body of work in economics on the persistence of historical shocks (Nunn, 2009, 2014; Spolaore and Wacziarg, 2013; Abramitzky, 2015; Michalopoulos and Papaioannou, 2017). The central question in this literature is not whether history matters in a descriptive sense, but through which mechanisms specific historical events continue to shape economic behaviour long after the original shock has passed. Along these lines, our conceptual setting brings together four broad ideas that have usually been studied separately. First, wartime violence may leave long-lasting local legacies even after physical reconstruction is complete. Second, preferences and beliefs may be shaped by past experiences and then show some persistence over time. Third, collective memory is not merely inherited: it can be politically produced, maintained, and periodically reactivated. Fourth, these forces can affect private market choices, including the national origin of goods purchased by households.

More specifically, the first strand of literature we were referring to studies the long-run effects of war and political violence. A central finding in this work is that material destruction need not translate mechanically into persistent economic underdevelopment. Miguel and Roland (2011), for example, study the long-run effects of U.S. bombing

in Vietnam and find little evidence of permanent local economic scarring in poverty, consumption, literacy, or infrastructure. Similarly, [Gagliarducci et al. \(2020\)](#) show that BBC radio reception during the Italian civil war increased partisan activity in the short run, but did not produce persistent differences in postwar political outcomes. These results are important because they discipline the interpretation of historical shocks: violence and wartime mobilization may have large immediate effects, but their long-run consequences are not automatic.

Our paper starts from this insight but shifts attention from physical destruction to the memory of war atrocities. The absence of permanent material scarring does not imply the absence of persistent behavioural effects. A province may rebuild its infrastructure, recover economically, and reintegrate into national and international markets while still retaining a politically meaningful memory of occupation, reprisals, and civilian victimization. In this sense, our analysis is complementary to studies that find limited long-run effects of wartime destruction. In fact, we ask whether the memory of violence, rather than the destruction itself, can affect market behaviour decades later.

The paper also relates to the broader literature on historical persistence and cultural transmission. [Nunn and Wantchekon \(2011\)](#) show that exposure to the slave trades is associated with lower levels of trust in Africa today, while [Voigtländer and Voth \(2012\)](#) document striking local persistence in anti-Semitic attitudes in Germany from the medieval period to the twentieth century.³ These contributions establish that historically rooted attitudes can persist for a very long time. Yet they also raise a further question: What keeps a historical attitude alive? Persistence may reflect intergenerational transmission within families, local social interactions, institutional reinforcement, or repeated political activation ([Bisin and Verdier, 2023](#)). Our contribution is to focus on the latter channel. In our setting, wartime violence becomes economically relevant not simply because it happened, but because a mass political organization (i.e., the main party on the left), kept the memory of Nazi-fascist violence salient in local public discourse and in local social life.

This mechanism connects our paper to research on the endogenous formation of

³See also [Becker and Pascali \(2019\)](#) along the same line of reasoning.

preferences and beliefs. [Stigler and Becker \(1977\)](#) argue that apparent changes in tastes can arise when past experiences alter the shadow prices associated with future choices. In our context, wartime violence can be interpreted as creating a stock of “memory capital” that raises the subjective cost of consuming German goods. This stock, however, need not depreciate at a natural rate. Political parties, civic organizations, commemorations, party newspapers, local rituals, and public narratives may maintain it over time.⁴ [Fehr and Hoff \(2011\)](#) emphasize that social institutions can sustain preferences by activating identities and norms. [Alesina and Fuchs-Schündeln \(2007\)](#) show that political regimes can shape preferences for state intervention, and [Malmendier and Nagel \(2011\)](#) show that lifetime macroeconomic experiences affect risk-taking and financial choices.⁵ These papers provide a general prior: preferences are not only fixed primitives, but are shaped by lived experience and institutional environments.

The same logic applies to attitudes toward foreign countries and foreign goods. [Guiso et al. \(2009\)](#) show that bilateral trust and cultural perceptions help explain trade, portfolio investment, and foreign direct investment between European countries, even after controlling for standard economic determinants. [Morey \(2016\)](#) similarly emphasizes that historical and cultural affinities contribute to home bias in international trade. Our paper provides a product-market and subnational account of one such cultural bias. Rather than inferring the importance of history from cross-country correlations in trade, we give an account of a specific historical shock, a clear political mechanism, and a specific market outcome: German passenger-car purchases across Italian provinces in the postwar decades.

A related literature shows that consumer tastes can persist even when individuals move across markets or when the original constraints that shaped choices have disappeared. [Atkin \(2013\)](#) shows that migrants in India continue to consume foods associated with their region of origin, even when those foods are relatively expensive in their destination. The persistence of food tastes is strongest for preferences formed early in life,

⁴The concept of “memory capital” is useful here because it clarifies the role of the PCI in our conceptual framework. The party is not assumed to create the original shock: Nazi-fascist violence did that. Rather, the party acts as an institution that maintains and refreshes the stock of memory associated with that shock.

⁵See also [Alesina and Giuliano \(2015\)](#) for a broad and in-depth discussion of the issue.

suggesting that consumption habits can become durable components of identity. Our setting differs in both the object and the channel. The relevant preference is not for a familiar regional good, but against goods associated with a clearly identified former occupying power. Moreover, the persistence we study is not only individual or familial. It is mediated by local political organization: provinces with stronger PCI support are precisely those in which the memory of wartime violence is more likely to be publicly commemorated and politically framed.

As we anticipated in the Introduction, the paper most closely related to ours is [Fouka and Voth \(2023\)](#), who study the effect of German wartime atrocities on Greek consumer behaviour during the European debt crisis. They show that Greek municipalities more exposed to German reprisals reduced purchases of German cars more sharply when the disagreement and friction between the Greek and German governments became politically salient after 2010. Their evidence demonstrates that collective memory can shape private market choices when an external event reactivates historical grievances; ours documents a persistent effect sustained across four postwar decades and ten national elections, without any external shock to reactivate it, and confined to household consumption goods rather than business purchases. Thus, our paper differs in a key respect. We do not study a dormant “memory capital” awakened by a later crisis. Instead, we study whether memory can operate continuously during ordinary peacetime when a political alignment keeps it alive. The distinction is central. In [Fouka and Voth \(2023\)](#), the crisis is the trigger; in our setting, the Italian Communist Party is the carrier. We also provide a sharper test of the sentiment channel: the concentration of the effect on authentically German-owned brands, and its attenuation when American-owned Ford and Opel vehicles manufactured in Germany are included, adds a layer of identification that their setting, focused on a single crisis episode, does not allow.

This emphasis on carriers of memory also links the paper to work on political communication and ideological transmission. [Cantoni et al. \(2017\)](#) show that school curricula can causally shape students’ political attitudes, while [Bursztyn and Cantoni \(2016\)](#) studies whether exposure to Western television affected consumption behaviour

in East Germany. These papers suggest that political and informational environments can affect private beliefs and choices. Our setting differs because the relevant institution is not the school curriculum or mass media exposure, but a national party organization embedded in local postwar political life. The PCI had both the incentive and the organizational capacity to transform wartime suffering into a persistent political narrative: anti-fascist resistance, victimization by Nazi-fascist violence, and suspicion toward West Germany as part of a broader Cold War discourse.⁶

Our paper also relates to recent work on Italian wartime violence and postwar politics. [Gagliarducci et al. \(2020\)](#) use variation in BBC reception to study partisan resistance and political mobilization, while [Bellodi et al. \(2024\)](#) use exposure to Nazi-fascist atrocities from the Atlas of Nazi-Fascist Massacres (i.e., our same source of data on violence) to study later political behaviour. We use the same historical setting to study a different outcome: consumer choice. This matters because consumption provides a distinct test of “memory capital”. Voting may directly reflect ideology, party organization, or local political networks. Car purchases, by contrast, are costly private decisions involving quality, price, availability, and brand reputation. Finding an effect in this domain therefore suggests that collective memory reached beyond formal politics into household choices.

A further connection is with the recent economic literature on fascism,⁷ its origins, and its long-run political and social legacies. Recent work has treated fascism not only as an ideological phenomenon, but as an outcome of local economic conflict, wartime disruption, elite reaction, and institutional mobilization. [Acemoglu et al. \(2022\)](#) show that the rise of fascism in Italy was closely linked to the post–World War I surge in socialist support: areas more exposed to WWI casualties experienced stronger socialist mobilization, which in turn generated a more pronounced violent fascist reaction, financed by elites threatened by redistribution. Complementing this work on fascism’s origins, [Carillo \(2022\)](#) studies Mussolini’s *Città di Fondazione* (“foundation cities”; i.e.,

⁶This does not require a connection between voters’ ideological positions and consumption preferences. The mechanism may operate through social norms, family narratives, local commemorations, and the political meaning attached to German brands. The empirical prediction is therefore straightforward: in provinces where casualty exposure was greater and PCI strength higher, German-branded cars should be a lower share of foreign-car purchases.

⁷See [Gentile \(2022\)](#) for a broad and comprehensive history of fascism.

the new towns built on reclaimed land as showcases of Fascist modernization) and shows that infrastructures created by the dictatorship can leave persistent political traces after democratization, increasing support for neo-fascist parties in the long run. Focusing on the same geographical area [Belmonte \(2020\)](#) and [Docquier et al. \(2026\)](#) study the educational and efficiency consequences of the resettlements in the Pontine Marshes. These papers are useful for our setting because they interpret fascism and its aftermath as spatially heterogeneous and institutionally persistent. Our paper studies a different but related legacy. Rather than focusing on the origins of fascist support, neo-fascist voting, or contemporary populism, we ask whether the experience of Nazi-fascist violence during 1943–45, politically preserved by the anti-fascist left, shaped private consumption choices in the decades after the dictatorship had collapsed.⁸

To summarize, our contribution to the existing literature is threefold. First, we document long-run peacetime persistence in a private consumption market. The effect we estimate spans four postwar decades and does not require a later crisis to become visible. Second, we identify an institutional mechanism: the interaction between local exposure to Nazi-fascist violence and the electoral strength of the PCI. This mechanism turns a historical wound into a durable consumer signal. Third, we provide evidence that the effect operates through household taste rather than supply disruption or local economic conditions. The effect is concentrated among German-owned passenger-car brands, is not mirrored by a decline in the overall foreign-car share, and is absent for commercial trucks. These patterns are difficult to reconcile with generic demand shocks, protectionist sentiment, or postwar reconstruction constraints. They are instead consistent with a national-origin bias sustained by politically mediated collective memory.

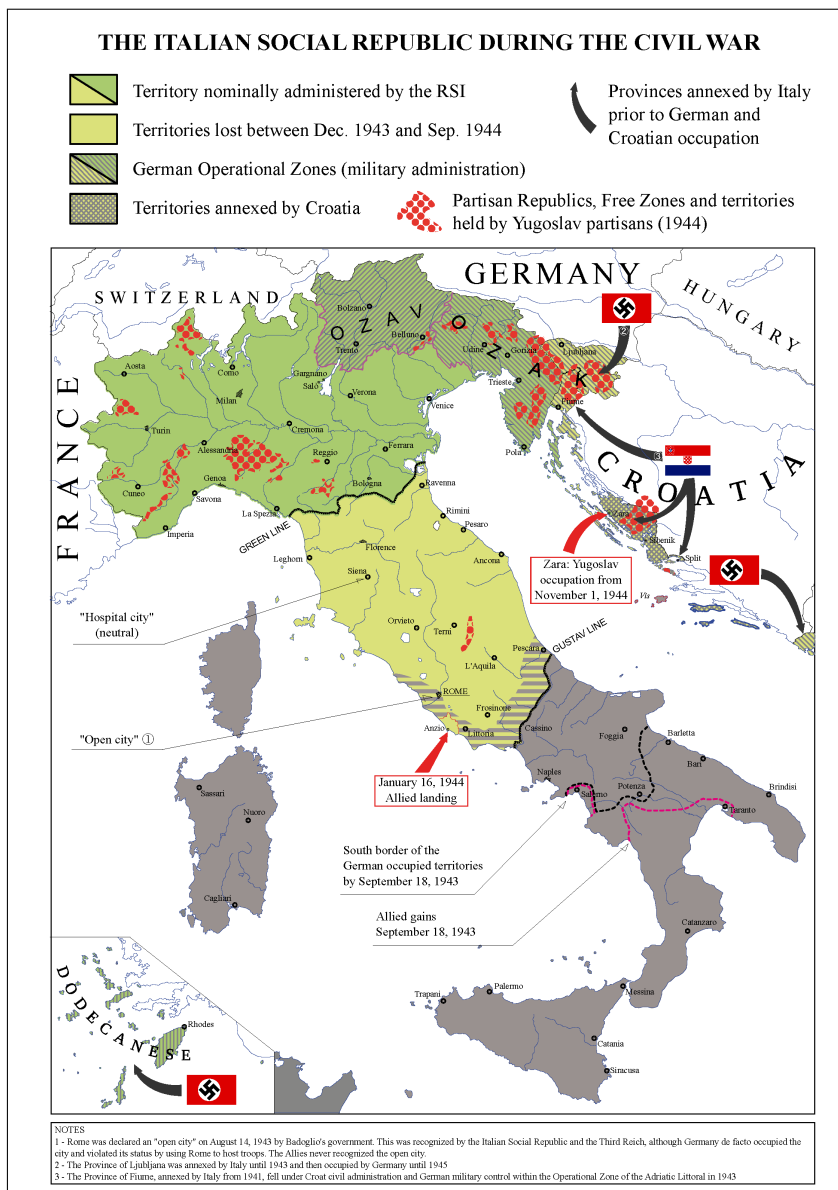
Lastly, by documenting the long-run economic consequences of Nazi-fascist violence, this paper contributes to keeping alive the memory of its victims.

⁸In this sense, the paper connects two sides of the legacy of fascism: the persistence of political cultures associated with fascism and neo-fascism, and the persistence of anti-fascist memory in places where Nazi-fascist violence was locally salient. Our empirical focus is on the second channel.

3 Brief historical background

The armistice of 8 September 1943 ended Italy's participation in the Axis and triggered an immediate German occupation of the Center and North of the country. Mussolini, deposed and imprisoned since late July, was reinstated as head of the *Repubblica Sociale Italiana*, a Nazi-backed puppet state entirely dependent on Berlin in military, economic, and political matters.⁹

Figure 1: The Italian Social Republic and German occupation zones, 1943–1945



⁹In the words of Rudolf Rahn, Germany's plenipotentiary to the RSI: "Everything in occupied Italy must be exploited by us for our war effort" (Holland (2008), page 111).

The military geography of the occupation was determined by two successive German defensive lines. The Gustav Line, running roughly east–west south of Rome, halted the Allied advance from December 1943 until May 1944. After its breakthrough, the front shifted north to the Gothic Line, a fortified barrier extending from La Spezia on the Tyrrhenian coast to Pesaro on the Adriatic, which the Germans held from the summer of 1944 until April 1945. As a result, the Center-North of Italy (the areas above the Gustav Line) remained under German occupation for periods ranging from under a year in the immediately northern provinces to nearly two years in Piedmont and Liguria (Fontana et al., 2023). The South, liberated rapidly by the Allies, experienced only a brief and comparatively less destructive occupation. Figure 1 illustrates the territorial organization of the occupation, showing the location of the defensive lines and distinguishing between the German Operational Zones under direct military administration (i.e., OZAV and OZAK zones) and the broader RSI-administered territory.

The occupation imposed severe material and human costs on the civilian population. In the territory under their control, the Germans requisitioned industrial equipment, seized food and raw materials, imposed forced labour, and deported workers and political prisoners to the Reich.¹⁰

3.1 Civilian casualties from political violence

The German army and its RSI allies carried out systematic violence against the civilian population. The motivations were multiple: reprisals for partisan attacks, the enforcement of territorial control, anti-Semitic persecution, and, in the later phases of the conflict, increasingly indiscriminate repression driven by the mounting pressure of the Resistance (Gagliarducci et al., 2020). Following Field Marshal Albert Kesselring’s June 1944 order, promising soldiers immunity from prosecution for acts of “excessive

¹⁰An estimated 40,000 civilians were deported to Germany. Of these, 6,806 were Jews, of whom 5,969 (nearly 88%) perished; in total, the *Centro di Documentazione Ebraica Contemporanea* (CDEC) has identified 7,579 Jewish victims of the Shoah in Italy for the period 1943–1945 (Centro di Documentazione Ebraica Contemporanea (CDEC), 2024). The victims of the deportations were not only Jews but also homosexuals, Jehovah’s Witnesses, Roma and Sinti, political opponents (Berenbaum, 1990). In the zones closest to the front, German forces also systematically razed villages and displaced populations to clear the terrain. Around 650,000 Italian disbanded soldiers (Italian Military Internees *Internati Militari Italiani* (IMI)) were also captured and deported to Nazi concentration camps after September 8, 1943. The RSI collaborated closely with these measures.

restraint”, attacks on civilians and communities became more frequent and more severe.

The scale and geography of this violence are documented in the Atlas of Nazi-Fascist Massacres (ANPI-INSMLI, 2016; Cacciatore and Pizzirusso, 2026), a comprehensive joint German-Italian research project that records 5,910 episodes of violence between July 1943 and May 1945, totalling more than 25,000 civilian casualties. The distribution across Italy was far from uniform. Casualties were heavily concentrated in the Center-North, particularly in the areas that lay close to the Gothic Line and that experienced the longest occupation.¹¹

The episodes also ranged widely in scale. The great majority were relatively small (i.e., the median event involved a single victim while the average was four), but a number of massacres claimed hundreds of lives and left lasting marks on collective memory: the Fosse Ardeatine in Rome (335 victims, March 1944), Sant’Anna di Stazzema in Lucca (392 victims, August 1944), and Marzabotto in Bologna (769 victims, September–October 1944). For an overview, Figure 2 illustrates the geographic distribution of violence across Italy, together with the positions of the Gustav and Gothic Lines, and the OZAV/OZAK zones. Each circle represents an event with the size proportional to the number of casualties in each episode.

3.2 The Resistance and the postwar role of the PCI

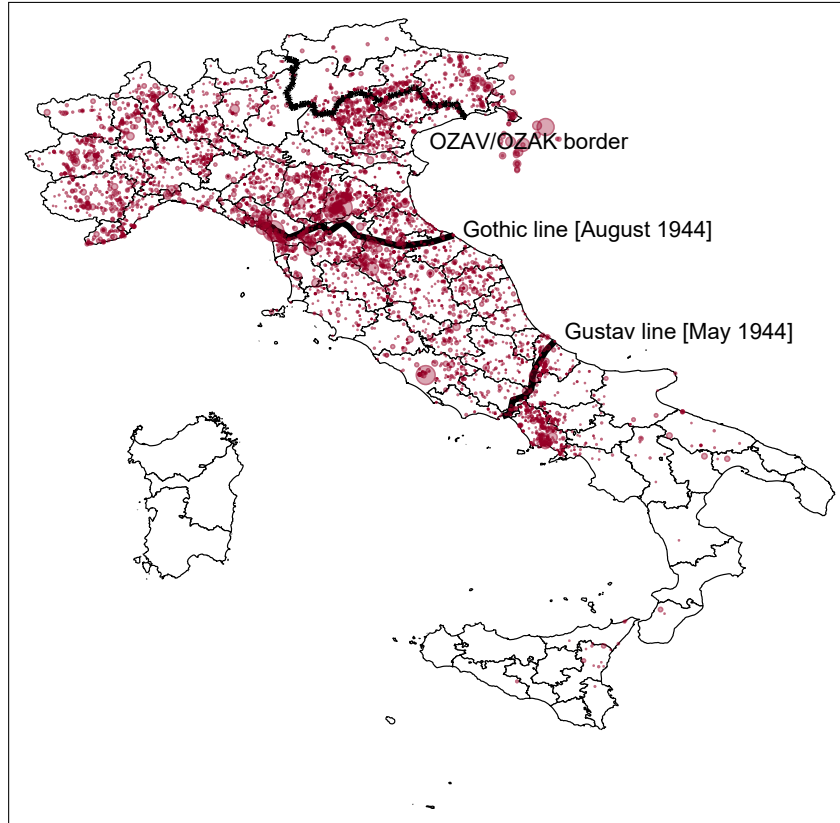
The conflict that followed took the form of a civil war: an organized armed movement (*Resistenza*) that grew from a few thousand fighters in the autumn of 1943 to tens of thousands one year later, supported by an estimated 20,000 civilians directly connected to the movement (Gagliarducci et al., 2020). The partisan brigades were organized along political lines. The largest and most active formations were left-wing, although Catholic, liberal, monarchist, and other groups also participated.¹² The Italian Communist Party played a leading organizational role throughout, and its cadres were disproportionately present both in the partisan leadership and in the civilian networks

¹¹The South and the two main islands were largely spared, having been liberated early by the Allied advance.

¹²The communist *Garibaldi* and the socialist *Matteotti* and *Giustizia e Libertà* brigades were the largest formations, together accounting for the majority of partisan fighters in the field. Catholic and monarchic groups were also present.

that sustained the Resistance.¹³

Figure 2: Geography of political violence



Notes: Each circle represents an episode of violence, with size proportional to number of casualties. Source: ANPI-INSMLI (2016).

Liberation on 25 April 1945 was followed by a rapid political consolidation. In the immediate postwar elections in 1948, the PCI performed significantly better in municipalities where the Nazi occupation had been longer and harsher, reflecting both the organizational legacy of the partisan brigades and the political salience of wartime suffering (Fontana et al., 2023). In subsequent decades, the PCI became the dominant opposition force in Italian politics, drawing its core electoral support from the so-called “red belt” of Emilia-Romagna, Tuscany, and Umbria (Fontana et al., 2023), the regions where the Resistance had been most active and wartime violence most intense.

The postwar role of the PCI as an active keeper of wartime “memory capital,” and

¹³The PCI had been founded in January 1921 in Livorno as a split from the socialist movement (Pasquino, 2009). Its strong organizational capacity in the factory and countryside, combined with its anti-fascist credentials from two decades of clandestine opposition, gave it a decisive advantage in building the Resistance network (Fontana et al., 2023).

the mechanisms through which it implicitly sustained anti-German sentiment across the provincial population, are examined in detail in the following section.

4 The PCI as a keeper of wartime memory

The historical facts established in the previous section do not by themselves generate a long-run effect on consumer preferences. The critical ingredient is a mechanism through which the memory of wartime violence was kept politically and socially active for decades after 1945 (see Figure A1 in the Appendix for two examples of election posters that visualize the concept). This section elaborates two features of the argument: the specific channels through which wartime memory was sustained and the population-level scope of the mechanism.

4.1 How the PCI kept wartime memory politically alive

The PCI's sustained effort to keep the memory of Nazi-fascist violence politically active operated through several reinforcing channels.

Annual commemorations of the Liberation on 25 April were the most visible. Rather than treating April 25 as a purely civic or memorial occasion, the PCI consistently organized and framed these events as explicitly political ones: occasions to reaffirm the party's anti-fascist identity, honour the partisan legacy, and situate wartime suffering within an ongoing political struggle. In provinces with strong support for PCI, these were major public events, drawing large turnouts and receiving extensive coverage in the local and national party press.

The largest massacres were central to this commemorative infrastructure. Major massacres such as the Fosse Ardeatine, Marzabotto, and Sant'Anna di Stazzema, were not merely remembered in the abstract: they were annually re-enacted as political reference points, with party officials and survivors' associations jointly leading public ceremonies that explicitly named the Nazi-German perpetrators and framed the violence as a national wound requiring sustained political response. In the areas hosting these events, the yearly commemoration cycle ensured that wartime violence remained part

of the active public discourse rather than fading into historical memory.

Anti-German sentiment was further sustained by the PCI's broader geopolitical positioning. The party's alignment with the Soviet bloc and its sustained opposition to NATO gave hostility toward Germany a contemporary political anchor: Germany was not only the country of the wartime perpetrators but also a founding member of the Atlantic alliance that the PCI sought to undermine. This framing kept anti-German feeling politically alive in the language of current affairs, rather than confining it to the domain of historical grievance. Italy's postwar economic recovery and reintegration into the European economy would otherwise have provided powerful countervailing forces favouring normalization of Italian–German relations.

Finally, the party's organizational apparatus provided a dense infrastructure for memory transmission in high-PCI provinces: *l'Unità* (“The Unity”; i.e., the party newspaper) and regional party publications, the mass popular gatherings of the *Feste de l'Unità*, ANPI events, and an extensive network of local party sections (*sezioni*).¹⁴ In these areas, the party press was widely read, ANPI sections were active and well-attended, and the local party section served as a community hub where wartime narratives were regularly recalled and politically interpreted. This organizational density meant that the commemorative function of the PCI was not confined to a few annual occasions but was woven into the fabric of everyday political life in the provinces where the party was strongest.

4.2 A provincial public discourse, not a partisan boycott

It is important to clarify and emphasize that we are not arguing that PCI voters individually chose not to purchase German cars as an act of partisan loyalty or political expression. The mechanism is not one of individual political behaviour but of collective

¹⁴The pervasiveness of this “memory capital” in Italian popular culture is illustrated by the hugely successful *Don Camillo* film series (1952 onward), based on Giovanni Guareschi's novels. Set in a fictional Po Valley village, the films centre on the comic rivalry between Don Camillo, the local Catholic priest, and Peppone, the PCI mayor and former partisan, whose wartime experience as a Resistance fighter shapes his entire postwar identity. That a conservative author writing for a mass audience across the political spectrum chose the partisan past as the defining trait of his communist character speaks to how thoroughly the memory of Nazi-fascist violence had become embedded in postwar Italian culture.

exposure to a sustained local public discourse.

In areas where the PCI was electorally strong, the entire local public environment (e.g., commemorative events, local media, civic associations, social networks) was permeated with narratives of wartime violence and anti-German sentiment. This exposure affected all residents of those areas, regardless of their own voting behaviour or political sympathies. They were all nonetheless exposed to the same annual April 25 commemorations, the same local newspaper coverage of massacre anniversaries, and the same social conversations shaped by a community in which wartime memory was politically salient. The mechanism is therefore one of local information environment effects rather than individual political identification: what matters is not whether a person voted Communist but whether they lived in a province where the PCI’s organizational strength generated a sufficiently dense and sustained public discourse around wartime violence.

This distinction has direct implications for the interpretation of our empirical analysis. The vote share for PCI captures the intensity of the party’s local presence and organizational capacity, and hence the intensity of the memory-keeping function it performed, rather than the proportion of the population that was politically sympathetic to its positions. A province with a PCI vote share of 40 percent was not a province where 40 percent of residents harbored anti-German sentiment for partisan reasons; it was a province where the PCI’s organizational infrastructure was strong enough to sustain a public commemorative culture that reached well beyond the party’s own electorate. An interaction term between PCI vote share and casualty exposure, instead, captures the extent to which this infrastructure found genuine historical material to work with: the memory-keeping function was more potent, and presumably reached a wider audience, in localities where the underlying wartime experience gave it greater credibility and emotional resonance.¹⁵

¹⁵More broadly, the narrative that blamed the massacres exclusively on Nazi brutality went in parallel with the construction of the myth of “the good Italian” (*il bravo italiano*; Focardi, 2023). This was a widespread phenomenon nationwide, instrumental for the reconstruction of a national identity in post-fascist republican Italy (Focardi and Klinkhammer, 2004).

5 Data

Our analysis draws on three main data sources: records of civilian casualties in 1943-45, vehicle registration data disaggregated by province and country of origin, and electoral results in national elections. We describe each in turn, along with the other variables used in the estimations. We begin by clarifying the unit of observation and the construction of our geographical units.

The empirical analysis is conducted at the province level, the most disaggregated unit at which vehicle registration data are available. This requires addressing two sources of boundary change: shifts in the number and borders of provinces during our sample period, and differences relative to the 1936 census definitions from which we draw several control variables.

We use the 1936 provincial boundaries as our benchmark throughout.¹⁶ The 1936 census recorded 94 provinces, three of which (i.e., Fiume, Pola, and Zara) were ceded to Yugoslavia after World War II and are excluded from our dataset. Of the remaining 91, Trieste presents a special case: it was administered as an international territory from 1947 to 1954 and returned to Italy with substantially reduced boundaries, having lost much of its peripheral and rural territory.¹⁷

Four further provinces (i.e., Caserta, Isernia, Oristano, and Pordenone, created in 1945, 1970, 1974, and 1968 respectively) were carved out of existing ones after 1936.¹⁸ We treat their territories as remaining part of their parent provinces (i.e., Naples, Campobasso, Cagliari, and Udine, respectively). Thus, our dataset covers 91 geographical units, with Trieste entering the sample after 1954.

5.1 Political violence

Our measure of political violence draws on the Atlas of Nazi-Fascist Massacres ([ANPI-INSMLI, 2016](#)), introduced in the previous section and discussed in [Cacciatore and](#)

¹⁶For provinces that changed name, we use the current one (i.e., Aquila instead of Aquila degli Abruzzi, Latina instead of Littoria, Taranto instead of Ionio, Udine instead of Friuli).

¹⁷Since we cannot reconcile the 1936 census data with the post-1954 configuration, we exclude Trieste in a robustness check.

¹⁸Caserta existed until 1927 when it was dismantled and then re-established in 1945.

Pizzirusso (2026). The Atlas records each episode with precise date, location, and information on victims (e.g. gender, age) and motivation (e.g. retaliation, territorial control).¹⁹ Since our outcomes are observed from the 1950s onward, we abstract from the time dimension and aggregate all episodes over the 1943–1945 period at the province level, exploiting cross-sectional variation in exposure.

Summary statistics are reported in Panel A of Table 1. There is substantial variation across provinces in both the number of events and total casualties, as reflected by the large standard deviations. Some of this heterogeneity reflects differences in province population, so we also report per-capita measures. Variation remains considerable even after this normalization, and it is somewhat reduced when considering the average number of casualties per event (around 4), which proxies for the severity of individual episodes rather than their cumulative toll. Similarly, a log version also smooths out the skewness of the distribution. It remains an open empirical question which measure best captures the effect on collective “memory capital,” as the salience of violence may respond differently to its intensity than to its aggregate scale. The substantial cross-province variation in both the incidence and the intensity of violence provides the key source of identifying variation in the empirical analysis.

5.2 Vehicle registrations

Vehicle registrations in Italy have been administered by ACI since its creation in 1905. The long continuity of ACI oversight ensures consistency across our sample period. The ACI annual yearbooks, now digitized and available through the ACI Digital Library (<https://bibliotecadigitale.aci.it/>), record registrations by vehicle type and province. For our purposes, the key feature is that they report both the country of origin and the vehicle brand (see Figure A3 in the Appendix for an example). Brand information is essential because country of manufacture and brand nationality do not necessarily coincide: Ford and Opel, for instance, operated production plants in Germany during our sample period while remaining American-owned brands. We also observe imports of Ford and Opel cars coming directly from the United States or from

¹⁹Not all details are available for each event.

Table 1: Summary statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Panel A: <i>political violence</i>					
Events	91	64.67	75.72	0	450
Casualties	91	268.27	390.70	0	2,747
Events per capita	91	0.16	0.16	0	0.74
Casualties per capita	91	0.65	0.86	0	4.95
Casualties per event	91	3.93	3.78	0	26.00
$\ln(1 + \text{Casualties})$	91	4.39	2.08	0	7.92
Panel B: <i>car registrations</i>					
Total	909	13,387.03	22,698.50	100	262,934
Italian cars	909	8,454.00	14,430.50	96	156,888
Foreign cars	909	4,933.03	9,483.19	0	125,359
Cars from France	909	1,927.24	3,388.89	0	38,636
Cars from Germany	909	1,977.30	3,845.95	0	54,102
Cars from Germany, excl. Ford & Opel	909	1,159.31	2,381.67	0	34,909
Cars from UK	909	202.96	471.20	0	6,614
Cars from USA	909	9.22	25.27	0	315
Panel C: <i>truck registrations</i>					
Total	894	922.18	1,317.78	40	16,061
Italian trucks	894	583.92	808.94	40	8,504
Foreign trucks	894	338.26	631.63	0	9,310
Trucks from France	894	74.94	146.52	0	2,123
Trucks from Germany	894	146.10	223.13	0	2,618
Trucks from Germany, excl. Ford & Opel	894	72.85	108.44	0	1,444
Trucks from UK	894	30.17	61.66	0	859
Trucks from USA	894	1.77	15.89	0	465
Panel D: <i>electoral data</i>					
Vote share of Italian Communist Party	909	0.26	0.11	0	0.56

Notes: per-capita variables calculated out of 1,000 inhabitants; truck variables refer to trucks used for goods transport.

the United Kingdom.

This situation raises the question of how Italian consumers in the 1950s–1990s perceived these brands, as either American or German. The Ford name is clearly associated with the United States: the name, the Model T, and Detroit are synonymous with American industrial identity. Opel is a more interesting case. The firm was founded in Germany in 1862 and began producing automobiles in 1899; it was acquired by General Motors (GM) in 1929 and remained a wholly owned GM subsidiary throughout our sample period. Whether Italian consumers perceived it as German or American

is therefore an open question, and one our data cannot resolve directly. To shed light on this question, we draw on the advertising campaigns of the time. In the late 1950s, Opel sold its flagship models in Italy with the explicit dual tag line *la vettura tedesca della General Motors* (“the German car by General Motors”), invoking both identities simultaneously, on cars whose large, chrome-heavy styling was itself unmistakably transatlantic (see Panel A of Figure A4 in the Appendix). By the early 1960s the German qualifier had disappeared; the Kadett was now advertised simply as *prodotto della General Motors* (“product of General Motors”; see Panel B of Figure A4 in the Appendix). By the late 1960s, one celebrated campaign surrounded the car with ice hockey players (see Panel C of Figure A4 in the Appendix), a sport clearly associated with North American culture rather than Germany. Through the 1970s and 1980s neither Germany nor GM featured prominently, replaced by a stateless brand identity and Italian lifestyle imagery. The endpoint of this arc was the 1994 Corsa campaign: not merely an advertisement featuring James Dean but a dedicated James Dean edition model (i.e., Opel Corsa “Jimmy” was a special limited edition), with the face of America’s most iconic postwar rebel and no reference to Germany whatsoever (see Panel D of Figure A4 in the Appendix). Opel’s Italian marketing, in short, progressively downplayed the brand’s German origins in favour of first an American corporate identity and then a cosmopolitan statelessness. This advertising evidence suggests that Italian consumers are more likely to have perceived Opel as an American rather than a German brand. The econometric results provide a revealed-preference test of this hypothesis: if consumers indeed treated Ford and Opel as non-German, excluding them from the dependent variable should strengthen the estimated effect of wartime violence on German car purchases.

From the ACI yearbooks, we compiled a dataset of new car and truck registrations²⁰ by province and country of origin for the election years between 1953 and 1991.²¹ Summary statistics are in panels B and C of Table 1. Among cars, Italian producers account

²⁰We refer to trucks used for goods transport. ACI distinguishes between trucks in general and *merci* (“freight”) trucks, but other freight vehicles (e.g., *motocarri*, three-wheeled goods vehicles) also exist. Non-freight trucks account for less than 10% of total trucks on average.

²¹National elections were also held in 1946 and 1948, but ACI yearbooks for those years do not report country of origin. For the 1992 election, we use 1991 registration data, as the 1992 yearbook does not record country of origin for cars and contains no truck data.

for more than 60% of the market on average. France and Germany are by far the most important foreign sources, each representing roughly 40% of imported cars. The UK accounts for just below 5% and the USA for even less, though records also exist for imports from Czechoslovakia, Japan, the Netherlands, Sweden, and the USSR. The table also reports registrations of cars produced in Germany but excluding Ford and Opel imported from there. Excluding these two brands substantially reduces the mean of German car registrations, by roughly 40%. Whether this distinction matters empirically is examined below. The data show substantial heterogeneity across provinces in both levels and composition: 3 provinces recorded no imports in 1953 (but all had some imports since 1963), and the average is skewed by large urban provinces, with the median of 2,516 foreign cars approximately half the mean.

Truck registrations (panel C) follow a broadly similar pattern.²² Italian producers again account for more than 60% of the market. Germany is the dominant foreign supplier at around 43%, roughly twice the French share, a contrast with car registrations where the two countries are at parity. As with cars, excluding Ford and Opel from German truck registrations results in a substantial reduction: the two American-owned brands account for approximately half of all trucks recorded as imported from Germany. Note though that while Germany dominates the foreign truck market, firms had access to vehicles from several other source countries (e.g., Japan, the Netherlands, Sweden, United Kingdom, United States, USSR).

5.3 Electoral data

Our sample covers the postwar national elections for which both vehicle registration data and comparable electoral results are available. The upper bound is set by the 1992 electoral reform, which replaced Italy's proportional representation system with a mixed system first applied in 1994; the last comparable election is therefore 1992. The lower bound is set by the ACI yearbooks, which do not record country of origin for 1946 and 1948. The resulting sample covers 10 elections with 91 provinces per year, except

²²The digitized ACI yearbook for 1976 does not cover the regions of Marche, Tuscany, and Umbria, resulting in a loss of 15 observations.

in 1953 when Trieste was not yet part of Italy.²³

The Italian parliament comprises a lower chamber (the *Camera*) and an upper chamber (the *Senato*). We focus on results for the lower chamber for two reasons. First, only individuals aged 25 or older could vote for the Senate, making its electorate older and more conservative. Second, Senate seats were allocated at the regional level, making outcomes less directly linked to the province level of our analysis. The *Camera* also features more seats (630 versus 315 for the Senate), allowing for finer provincial variation. For each province we calculate the PCI vote share in *Camera* elections.²⁴ Summary statistics are in panel D of Table 1 while Figure A5 in the Appendix shows province-level vote shares for each election alongside the cross-province average. Overall, the data reveal substantial heterogeneity: some provinces gave the PCI an absolute majority, while in others the party received very few votes and was absent altogether in some years in Aosta. Spatially, Emilia-Romagna and Tuscany consistently record the highest shares, forming the party’s electoral stronghold. Over time, the average vote share rises steadily to its peak in 1976, followed by a continuous decline through 1992.

5.4 Other variables

Most remaining control variables are drawn from the 1936 census, which provides province-level data on population, density, the share of the active population, and the sectoral composition of employment across agriculture, industry, trade, and services. Provincial GDP data are not available for this period, but we supplement the census with deposits held at credit institutions (from the 1937 Annual Statistical Yearbook) and illiteracy rates (from the 1931 census). Together, these indicators document the substantial structural heterogeneity across Italian provinces that our province fixed effects will be able to absorb.

We also draw on province-level data on Jewish deportations from the *Centro di Documentazione Ebraica Contemporanea (CDEC) (2024)*, which reports that 6,806 Jews were deported from Italy during 1943–1945. These data capture a distinct di-

²³The elections took place in 1953, 1958, 1963, 1968, 1972, 1976, 1979, 1983, 1987, and 1992.

²⁴The PCI was dissolved in February 1991 and replaced by the Partito Democratico della Sinistra (PDS), which ran in the 1992 elections.

mension of Nazi persecution from the casualties recorded in the Atlas: deportations targeted Jewish communities specifically, whereas the Atlas documents broader civilian violence. From several periodical publications of the Italian National Statistical Institute,²⁵ we derive the provincial-level emigration towards Germany, France, and Europe (and overall emigration flows). We include these flows as controls since exposure to postwar Germany through emigration and return migration (whether permanent or temporary) may have reduced anti-German sentiment, potentially attenuating the memory-based effect we identify.

Summary statistics for all these time-invariant controls are reported in Table A1 in the Appendix.

6 Econometric methodology

Our empirical strategy exploits cross-provincial variation in wartime casualties, interacted with the PCI vote share across ten national elections, to identify the long-run effect of wartime violence on German car purchases. Since casualties are time-invariant at the province level, their direct effect is absorbed by province fixed effects; it is only through the interaction with the time-varying PCI vote share that the effect can be identified. The mechanism sustaining this interaction was developed in Section 4: the PCI’s organizational strength translated provincial casualty exposure into a sustained public discourse hostile to German goods. As a result, we estimate the following type of specification:

$$Y_{ct} = \beta_0 + \beta_1 \text{PCI_share}_{ct} + \beta_2 \text{PCI_share}_{ct} \times \text{Casualties}_c + \alpha_c + \alpha_t + \epsilon_{ct} \quad (1)$$

where Y_{ct} is our outcome of interest (i.e. share of car or truck registrations, out of foreign or total registration) in province c in election year t , PCI_share_{ct} is the election share of the Italian Communist Party in year t , Casualties_c is the chosen measure of province-level casualties and α_c and α_t represent province and year fixed effects,

²⁵We used the Emigration Statistical Yearbook (*Annuario statistico dell’emigrazione*), the Migratory Movement of the Resident Population (*Movimento migratorio della popolazione residente*; the Yearbook of Demographic Statistics (*Annuario di statistiche demografiche*) and the Yearbook of Labour and of Emigration Statistics (*Annuario di statistiche del lavoro e dell’emigrazione*).

respectively. ϵ_{ct} is the error term and we cluster standard errors at the province level. Our key prediction is that $\hat{\beta}_2 < 0$: for a given level of casualties, provinces with stronger PCI electoral support should exhibit larger reductions in German car purchases.

In our benchmark specifications, the dependent variable is the share of cars (or trucks) from a given country of origin among total foreign registrations. The advantage of working with shares is that they implicitly control for province size, which province fixed effects do not fully absorb given the large demographic and economic changes that occurred in Italy over the postwar decades. We estimate Equation (1) by ordinary least squares, given the large number of province and year fixed effects, and consider fractional estimators as a robustness check.

The casualty measure Casualties_c can take several forms: total casualties in absolute or per-capita terms, number of casualties per event, or the log of casualties. We experiment with all of these to assess robustness and to shed light on which dimension of violence matters most for collective memory, since a single large massacre may leave a deeper mark than many smaller incidents with the same aggregate death toll (or vice versa).

The challenge in estimating Equation (1) is the possible endogeneity of PCI_share_{ct} , which may be correlated with unobserved province-level characteristics affecting vehicle demand. Electoral support for the PCI may proxy for local political preferences, socioeconomic conditions, or persistent ideological traits that are jointly related to consumption patterns and to the extent of wartime atrocities. This raises a concern of omitted variable bias, reinforced by the fact that members of the Resistance were predominantly drawn from left-leaning segments of the population. Several features of our design, however, mitigate this concern. Province fixed effects absorb all time-invariant provincial characteristics, so identification comes from within-province variation in PCI vote share over time. Anticipating some of our results, brand specificity (i.e., the finding that the effect concentrates on German-owned brands and is absent for American brands manufactured in Germany) is difficult to reconcile with a generic story of left-wing consumption attitudes, as is the absence of any effect for commercial trucks. Discretizing PCI strength into quartiles further reduces sensitivity to outlier provinces

with extreme political profiles.²⁶

7 Results

Before turning to the main empirical specification in Equation (1), we present motivating evidence based on the German occupation zones (see the discussion in Section 3). Since zone classification is determined by military geography rather than by the realized incidence of violence, it provides a conceptually distinct and simpler, if less precise, test of the mechanism. We then estimate Equation (1) with the various casualty measures, and examine a series of robustness checks. Next, we try to uncover the underlying forces at work.

7.1 Motivating evidence: occupation zones

This exercise serves both as a sanity check and as an intuitive illustration of our identification strategy: if wartime exposure shapes postwar consumer behaviour through the mechanism we propose, provinces subject to more intensive German military control should exhibit lower German car purchases, and the effect should be stronger where PCI electoral support was higher. Specifically, we replace **Casualties_c** in equation (1) with a set of zone dummies (**Zone_1**, **Zone_2**, **Zone_3**, with **Zone_0**, the unoccupied southern provinces, as the baseline) interacted with **PCI_share**. Columns (1) and (3) include region fixed effects but not province fixed effects, so the zone-level dummies themselves can be estimated. Columns (2) and (4) include province fixed effects, which absorb the time-invariant zone classification.

Table 2 reports the results for both outcome variables: the share of all German cars among total foreign cars (Y^{foreign}) in columns (1)–(2), and the share of German-branded cars among total foreign cars (Y^{foreign} , excluding Ford & Opel) in columns (3)–(4).²⁷

²⁶To address residual endogeneity, we plan to use an instrumental-variables strategy, relying on the evidence in Acemoglu et al. (2022) that footsoldier casualties during WWI and fascist vote shares in the 1921 or 1924 elections are strong predictors of subsequent PCI vote shares. This approach isolates the component of **PCI_share_{ct}** that is plausibly unrelated to contemporaneous determinants of vehicle purchases, yielding a causal estimate of the interaction between political salience and the legacy of wartime violence.

²⁷As discussed in Section 5, Ford and Opel were both American-owned brands. The former had an

Table 2: Motivating Evidence: Occupation Zones

	$Y^{foreign}$ (all)		$Y^{foreign}$ (excl. Ford & Opel)	
	(1)	(2)	(3)	(4)
Zone 1	0.083** (0.031)		0.099* (0.056)	
Zone 2	0.107** (0.041)		0.113* (0.063)	
Zone 3	0.094** (0.040)		0.207*** (0.066)	
PCI share $\times$ Zone 1	-0.350** (0.164)	-0.525** (0.245)	-0.278** (0.128)	-0.341* (0.196)
PCI share $\times$ Zone 2	-0.346** (0.147)	-0.471** (0.211)	-0.326*** (0.100)	-0.338** (0.138)
PCI share $\times$ Zone 3	-0.621*** (0.191)	-0.673*** (0.234)	-1.192*** (0.214)	-1.218*** (0.295)
Year FE	Yes	Yes	Yes	Yes
Region FE	Yes	No	Yes	No
Province FE	No	Yes	No	Yes
R^2	0.481	0.511	0.424	0.466
Observations	909	909	909	909
<i>Total effect of PCI vote: p-value</i>				
Zone 1	0.178	0.456	0.067	0.148
Zone 2	0.096	0.481	0.000	0.007
Zone 3	0.011	0.280	0.000	0.000

Notes: Zone 0, corresponding to unoccupied provinces, is the omitted category; standard errors clustered by province in parentheses with *, **, and *** denoting significance at the 10%, 5%, and 1% level, respectively.

The zone-level coefficients in columns (1) and (3) indicate that provinces in occupation Zones 1–3 registered higher average German car shares than the unoccupied South, a pattern consistent with the greater wealth and urbanization of northern and central Italy. The interaction terms, however, tell a strikingly different story: across all four specifications, a higher PCI_share is associated with a significantly lower German car share in provinces that experienced heavier German occupation. The bottom panel of Table 2 reports p-values for the test that the total effect of PCI_share within each zone is zero. This is where the difference between the two versions of the dependent variable becomes clear: when excluding Ford and Opel cars, the null hypothesis of no effect for Zone 2 and Zone 3 is rejected at the 1% level with either region or province fixed effects (and at conventional significance levels when province fixed effects are in-

unmistakable American association. As for Opel, the advertising campaigns systematically weakened their German origins.

cluded) while the equivalent effect is not statistically significant for all imports from Germany. In conclusion, the net association between PCI electoral strength and German car purchases is negative and significant precisely in the provinces most exposed to wartime violence and when isolating a specific anti-German sentiment.

7.2 Main results

We now turn to Equation (1), which focuses on province-level measures of civilian casualties interacted with the PCI vote share. Table 3 reports results for Y^{foreign} excluding Ford & Opel across five alternative casualty measures; the bottom row reports the p-values of the interaction term of each specification when all German imports are included in the dependent variable instead.

Column (1) uses the total number of **casualties** in absolute levels, and shows that the interaction coefficient is negative but not statistically significant, probably because of the large cross-province variation in population size: absolute casualty counts are mechanically larger in more populous provinces, conflating the scale of violence with the size of the affected population. Columns (2) and (3) address this concern by using **casualties** per capita and **casualties** per event, respectively. Both interaction terms are negative and statistically significant, with average casualties per event showing a stronger significance at 5% level. This measure may be more appropriate: for a given level of per-capita exposure, violence concentrated in fewer episodes is likely to be more salient and more amenable to collective commemoration than the same toll spread across many small incidents. The named massacres that became focal points of postwar “memory capital” (the ones we already mentioned: i.e., Fosse Ardeatine, Marzabotto, Sant’Anna di Stazzema) had high numbers of casualties per event. At the same time, very large massacres may be outliers whose influence on the estimates is disproportionate; we return to this concern in the robustness section, where we allow large events to have a differential effect. Column (4) replaces the continuous version with indicators for the second and third terciles of the distribution of **casualties** per event, allowing for non-linearities in the response. Both interactions are negative and statistically significant, with the third-tercile coefficient (-0.401) larger in absolute

value than the second (-0.332), consistent with an increasing relationship between the intensity of violence and its effect on consumer preferences. Column (5) uses the log transformation $\ln(1 + \text{casualties})$, yielding a negative and significant interaction coefficient that reinforces the case for non-linear effects.

Table 3: Casualties and German Car Purchases

	$Y^{foreign}$ (excluding Ford & Opel)				
	(1)	(2)	(3)	(4)	(5)
PCI share	-0.292 (0.177)	-0.215 (0.177)	-0.204 (0.194)	-0.013 (0.220)	0.015 (0.224)
PCI share $\times$ casualties	-0.000 (0.000)				
PCI share $\times$ casualties per capita		-0.148* (0.085)			
PCI share $\times$ casualties per event			-0.027** (0.012)		
PCI share $\times$ casualties per event 2 nd tercile				-0.332* (0.183)	
PCI share $\times$ casualties per event 3 rd tercile				-0.401** (0.156)	
PCI share $\times$ $\ln(1 + \text{casualties})$					-0.072** (0.034)
Year FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
R^2	0.461	0.463	0.462	0.464	0.463
Observations	909	909	909	909	909
<i>p-value of interaction term if including Ford & Opel:</i>					
	0.311	0.101	0.106	0.210 0.004***	0.021**

Notes: standard errors clustered by province in parentheses with *, **, and *** denoting significance at the 10%, 5%, and 1% level, respectively.

In all, the negative and statistically significant effect on German car purchases holds across four measures that control for “size”, indicating that our main finding is not an artifact of how wartime violence is quantified. Going forward, we consider casualties per event our preferred measure: it removes population-size confounding, captures the salience of concentrated violence that made individual episodes focal points of postwar “memory capital,” and yields the strongest interaction effect.

A natural internal consistency check is whether the negative effect concentrates on German-owned brands. If the mechanism operates through anti-German consumer sentiment, we would expect the effect to be absent for American-owned brands that

happened to be manufactured in Germany, as Italian consumers would not associate them with Germany. The bottom panel of Table 3 reports the p-values of the interaction term when the dependent variable is Y^{foreign} , which includes Ford and Opel cars. The p-values are generally larger, and the evidence is less systematically significant (i.e., 2 versus 5 statistically significant estimates when excluding Ford and Opel). This brand-specificity pattern is precisely what a consumer-sentiment channel predicts.

To assess the economic relevance of our estimates, Table 4 translates the coefficients from Table 3 into percentage changes in Y^{foreign} (excluding Ford & Opel) relative to its sample mean of approximately 27 percentage points. For each of the four significant casualty measures (i.e., we do not report the effect for total casualties), we compute the reduction in German car share associated with a one-standard-deviation increase in that measure, evaluated at four representative levels of **PCI_share**: the 25th percentile (0.18), the sample mean (0.26), the 75th percentile (0.32), and the maximum observed value (0.56). For the tercile specification in column (4) of Table 3, we instead report the reduction associated with moving from the first to the second and third tercile of the **casualties** per event distribution. The magnitudes displayed in the table are economically meaningful and consistent across specifications. At the sample mean of the **PCI_share**, a one-standard-deviation increase in casualties per capita reduces the German car share by approximately 11.9% relative to the mean. For average **casualties** per event and the log transformation, the corresponding reductions are 9.5% and 14.1%, respectively, which are very much comparable. The effects grow substantially at higher **PCI_share**: in provinces where the PCI received its maximum recorded vote (0.56), a one-standard-deviation increase in casualties per capita reduces the German car share by 26.2% of the mean. The tercile comparisons produce much larger estimates, and should be interpreted as capturing the full contrast between provinces at opposite ends of the intensity distribution rather than as a marginal effect comparable to the continuous specifications.

Table 4: Economic magnitudes

PCI share	casualties per capita	casualties per event	$\ln(1 + \text{cas.})$	2 nd vs. 1 st tercile	3 rd vs. 1 st tercile
25 th percentile (0.18)	−8.3	−6.6	−9.8	−21.9	−26.5
Mean (0.26)	−11.9	−9.5	−14.1	−31.3	−37.9
75 th percentile (0.32)	−14.9	−11.9	−17.7	−39.3	−47.5
Maximum (0.56)	−26.2	−20.9	−31.0	−68.9	−83.3

Notes: each cell reports the percentage change in the mean of $Y^{foreign}$ (excluding Ford & Opel) for the given PCI_share percentile and casualty measure. Calculations based on the estimates in Table 3.

7.3 Robustness

Before proceeding further to better understand the possible forces behind the result we find, it is important to ascertain that the result is robust. To this end, Table 5 reports estimates from five alternative specifications. Across all columns, our key interaction coefficient remains negative, statistically significant, and of broadly similar magnitude as in Table 3, with estimates ranging from -0.026 to -0.031 for the specifications directly comparable (i.e., not column (2)). As in Table 3, the bottom of the table reports the p-value of the relevant interaction term when all German imports (i.e., including Ford and Opel), are included in the dependent variable, serving as a further consistency check that the effect is concentrated on authentically German-branded vehicles.

In column (1), we consider a fractional logit model, given that our dependent variable is a share bounded between zero and one. We do not adopt this as the benchmark estimator given the large number of province and year fixed effects, but the results demonstrate that the choice of estimator does not drive our findings. The marginal effect of the interaction between **casualties** per event and PCI_share is -0.026 , which is indistinguishable from the OLS estimate of -0.027 in column (3) of Table 3.²⁸

Column (2) addresses the functional form of the PCI_share interaction. While Table 3 already examines non-linearities in the **casualties** measure, here we allow the effect of **casualties** to differ non-linearly across the PCI_share distribution by replacing the continuous vote share with quartile dummies. This specification also

²⁸The marginal effect would be -0.025 with a fractional probit specification, which is not surprising since any difference between probit and logit would depend on the number of observations near 0 and 1, which represent less than 5% of our sample.

Table 5: Robustness checks

	Y^{foreign} (excl. Ford & Opel)				
	(1) Fractional logit	(2) Non- linear PCI share	(3) Pre-war trends	(4) Large massacres	(5) Jews de- portations
PCI_share	-0.696 (1.089)		-0.090 (0.246)	-0.192 (0.196)	-0.126 (0.200)
PCI_share $\times$ casualties per event	-0.141** (0.058)		-0.027** (0.012)	-0.031** (0.013)	-0.030** (0.012)
PCI_share, 2 nd quartile		-0.019 (0.020)			
PCI_share, 3 rd quartile		-0.004 (0.023)			
PCI_share, 4 th quartile		0.006 (0.034)			
PCI_share, 2 nd q. $\times$ cas. per event		-0.004 (0.002)			
PCI_share, 3 rd q. $\times$ cas. per event		-0.005*** (0.002)			
PCI_share, 4 th q. $\times$ cas. per event		-0.007** (0.003)			
Large massacre $\times$ PCI_share				0.442 (0.955)	
Large massacre $\times$ PCI_share $\times$ casualties per event				-0.010 (0.118)	
PCI_share $\times$ Deported Jews p.c.					-1.129** (0.569)
PCI_share $\times$ casualties per event $\times$ Deported Jews p.c.					0.075** (0.034)
Pseudo R^2	0.053				
R^2		0.462	0.468	0.462	0.462
Year FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Observations	909	909	909	909	909
Marginal effect	-0.026				
<i>p-value of interaction term if including Ford & Opel:</i>	0.104	0.360	0.096*	0.088*	0.084*
		0.001***			
		0.759			

Notes: standard errors clustered by province in parentheses with *, **, and *** denoting significance at the 10%, 5%, and 1% level, respectively; pre-war province characteristics (agriculture share, industry share, illiteracy rate, population density, bank deposits) interacted with a linear time trend included in column (3).

partially mitigates endogeneity concerns, since it relies on a coarser classification of PCI strength rather than on the exact vote share, reducing sensitivity to any province-level unobservables that are correlated with the precise level of electoral support.²⁹ The interaction coefficients follow a monotone gradient, which is consistent with the continuous specification and suggests that the sentiment effect strengthens progressively with PCI organizational capacity. The pattern also indicates that a minimum threshold of party presence was needed for the memory-keeping mechanism to become precisely operative, as the second quartile produces a negative but imprecise estimate.

In the specification of column (3), we add interactions of pre-war provincial characteristics with a linear time trend, allowing provinces with different structural features to exhibit different secular trends in German car demand. In particular, we include the share of active population in agriculture and industry, population density, bank deposits per capita (all from 1936), and the illiteracy rate (in 1931). These additional controls are individually insignificant and leave the interaction coefficient essentially unchanged at -0.027 , confirming that the result is not an artifact of differential pre-war trends.³⁰

In column (4) of Table 5 we allow the effect to differ for the three provinces that experienced the largest individual events (Marzabotto in Bologna, Fosse Ardeatine in Rome, and Sant’Anna di Stazzema in Lucca) by introducing interaction with an indicator variable for these areas. These provinces might be outliers if the salience of a single catastrophic event operates through a different mechanism than the accumulated toll of smaller episodes. The double interaction between `PCI_share` and the large-massacre dummy is positive but very imprecisely estimated. The triple interaction that also includes the `casualties` per event (i.e., the focus of our analysis) is negative, suggesting a somewhat stronger effect in these provinces, but equally imprecisely identified (possibly due to the small number of affected provinces). In fact, a test that the total effect

²⁹Quartiles are computed on the pooled distribution of `PCI_share` across all province-year observations. Thus, the cutoffs are fixed but provinces can move between quartiles across elections as their vote share changes. Identification of the interaction terms therefore comes from both cross-sectional variation in average `PCI_share` strength and within-province changes over time. In practice, 89% of provinces switched quartile at least once over the sample period, so identification of the interaction terms comes predominantly from within-province variation over time. This within-province variation is less susceptible to bias from time-invariant unobservables than the continuous specification, providing a further attenuation of endogeneity concerns.

³⁰Replacing the linear trend with year fixed effects interacted with these pre-period controls yields a coefficient of -0.024 , significant at 5% level.

for large-massacre provinces is zero cannot be rejected (p-value = 0.734). Instead, the main interaction coefficient is essentially unchanged at -0.031 .

Finally, in column (5) we augment the benchmark with Jewish deportations per capita and its interactions with `casualties` per event and `PCI_share`. Deportations represent a distinct dimension of Nazi persecution targeting Jewish communities specifically, whereas the Atlas records broader civilian violence, regardless of possible racial motives; the two channels may independently contribute to anti-German sentiment. The coefficient on `PCI_share` interacted with deportations per capita is negative and significant at the 5% level, indicating that provinces with both stronger left-wing support and greater deportation exposure purchased fewer German cars. The triple interaction with `casualties` is positive and significant though its interpretation requires care. Since Jewish deportations were concentrated in urban areas while casualties from Nazi-fascist massacres occurred predominantly in more rural settings, the two variables capture largely distinct geographic experiences of Nazi persecution rather than overlapping ones. The positive triple interaction likely reflects this geographic separation: in provinces where deportations were high, `casualties` tended to be low, mechanically attenuating the `casualties` $\times$ `PCI_share` interaction. What matters for our purposes is that the benchmark `PCI_share` interaction remains negative and significant once both channels are controlled for, confirming that the main result is not confounded by the deportation channel.³¹

In conclusion, our key interaction coefficient is stable across different estimators, a non-linear specification of `PCI_share`, the inclusion of time-trend controls, allowing for heterogeneity across province types, and considering deportations.³² What is equally telling is the bottom row of Table 5: in every single column, folding American-owned Ford and Opel vehicles into the dependent variable causes the p-value to rise. This pattern, again, speaks to a finding that is neither a statistical artifact nor a generic

³¹To further investigate the urban versus rural possible divide, we estimated a specification with an interaction with an indicator variable for the provinces that include the administrative capital of each region (i.e., provinces are aggregated in twenty regions). It turns out that our key result vanishes for these provinces (results available upon request) and becomes slightly larger (in absolute terms) for the others.

³²The results are also robust to excluding Trieste, Naples (and both) on the basis that they experienced substantial boundary changes during or immediately after the war (results available upon request).

anti-foreign effect, but a precise and persistent aversion to goods carrying a German national identity.

7.4 Discussing the mechanisms

The main results of our analysis establish that higher wartime casualties, interacted with PCI electoral strength, reduced the share of German passenger cars among foreign car registrations. The PCI’s role as keeper of wartime “memory capital” provides the transmission mechanism, but the underlying nature of the channel remains to be established. We now discuss four pieces of evidence that together characterize it as a precise, taste-based aversion to goods carrying a German national identity, operating within the foreign segment of consumer demand and leaving business purchasing decisions untouched.

Brand specificity. First, the effect is concentrated on authentically German-branded vehicles: adding American-owned Ford and Opel cars manufactured in Germany but associated with the United States into the dependent variable systematically weakens and often eliminates the effect, as shown in the bottom panels of Tables 3 and 5 and also in the motivating evidence of Table 2. This result indicates that we do not have a generic aversion to goods manufactured in Germany, which would have also depressed demand for Ford and Opel, but a precise response to perceived national brand identity. This distinction already rules out supply-side explanations whereby German exporters were avoiding high-casualty markets, since such forces would operate on the basis of country of manufacture rather than brand nationality. It is also not the case that wartime violence reduced the propensity to import altogether, as we show in the next paragraph that the overall share of foreign cars among total registrations is unaffected.

Substitution patterns. The decline in German car purchases raises the question of where consumers redirected their demand. To answer this question, Table 6 reports estimates of Equation (1) using the shares of French, British, and American cars among total foreign cars, and the overall share of foreign cars among total registrations. Column (1) shows a positive but insignificant coefficient on `PCI_share` in the French car

regression, providing some indication that provinces with stronger left-wing support purchase a higher share of French cars among foreign vehicles. Coefficients for the interaction terms for British and American cars are negative but insignificant. Crucially, column (4) shows that the interaction of `PCI_share` with casualties has no effect on the overall share of foreign cars among total registrations. Thus, wartime violence did not reduce the propensity to buy imported vehicles but it altered the composition of imports.³³ This pattern is consistent with a two-stage decision process: consumers first choose between Italian and foreign vehicles, then select among foreign brands. The memory-based aversion operates within the second stage, leaving the first unaffected.³⁴

Table 6: Substitution patterns and emigration

	<i>Country's cars / Total foreign</i>			<i>Foreign / Total</i>	<i>Emigration</i>
	<i>France</i>	<i>UK</i>	<i>USA</i>		
	(1)	(2)	(3)	(4)	(5)
PCI share	0.585** (0.262)	-0.117* (0.062)	-0.137* (0.072)	-0.133* (0.053)	-0.174 (0.189)
PCI share × casualties per event	0.008 (0.016)	-0.002 (0.004)	-0.001 (0.003)	0.008 (0.005)	-0.039*** (0.012)
Emigration Germany p.c.					0.001 (0.004)
Emigration Germany p.c. × casualties per event					0.001*** (0.000)
Year FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
R^2	0.504	0.576	0.223	0.972	0.470
Observations	909	909	909	909	909

Notes: standard errors clustered by province in parentheses with *, **, and *** denoting significance at the 10%, 5%, and 1% level, respectively.

³³Vehicle registrations also record imports from Czechoslovakia, Japan, the Netherlands, Sweden, and the USSR. As each of these source country represents a small share of the market, we cannot estimate separate regressions for them. The null effect on total foreign share implies that the displaced German demand was redistributed across the range of available foreign brands rather than being absorbed by any single country in a precisely estimable way.

³⁴This two-stage interpretation is consistent with the nested demand structure common in the automobile literature. [Goldberg \(1995\)](#), in the US context, and [Goldberg and Verboven \(2001\)](#), for European car markets, both model automobile choice as differentiated along two dimensions: vehicle class and domestic versus foreign origin, and find empirical support for treating origin and class as distinct substitution groups. In particular, [Goldberg \(1995\)](#) defends the domestic versus foreign node based on IIA specification tests; [Goldberg and Verboven \(2001\)](#) report cross-price elasticities that are substantially higher within than across origin groups, a pattern especially pronounced in Italy. Our evidence sharpens this structure: the aversion we document operates within the foreign segment, targeting German-branded vehicles without affecting the overall propensity to purchase imports.

Emigration and attenuation. A complementary piece of mechanism evidence comes from postwar emigration flows. If the sentiment effect is driven by collective memory of wartime violence, direct personal exposure to postwar Germany through emigration should work in the opposite direction, partially eroding the aversion towards Nazi-Germany, if based on positive contemporary experiences. We test this hypothesis by augmenting the benchmark specification with (time varying) emigration to Germany per capita and its interaction with `casualties` per event. The results confirm this intuition: the interaction of `casualties` with emigration to Germany is positive and highly significant, indicating that provinces with greater postwar emigration to Germany experienced a partial attenuation of the memory-based aversion. Importantly, the main `PCI_share` interaction remains negative and increases (in absolute terms) to -0.039 , consistent with emigration having partially offset the sentiment effect in the raw data: once this channel is accounted for, the underlying memory-based aversion appears stronger.³⁵

The truck placebo. A key test of the consumer-sentiment interpretation is whether the effect extends to commercial trucks, which are purchased by firms for freight transport. Business procurement decisions are driven by price, reliability, and availability rather than by emotional associations with national identity. If the effect we uncovered reflects instead a supply-side disruption, we would expect similar results for trucks. Notice that the results on brand specificity already rule out supply-side disruptions but we provide a further test of this hypothesis in Table 7, which replicates the five specifications of Table 3 for the share of German trucks (excluding Ford and Opel) among total foreign truck registrations.³⁶ Across all five columns, the interaction term is small, changes sign across specifications, and is not statistically different from zero. The null

³⁵We also experimented by adding emigration to France and its interaction with casualties as a placebo (on top of emigration to Germany). In fact, French emigration provides postwar exposure to a foreign country that was not the source of the wartime violence, so its interaction with casualties should not attenuate the aversion to German goods. Indeed, the interaction term is positive and only marginally significant (i.e., p-value of 0.10) and leaves the German emigration interaction essentially unchanged, suggesting that the attenuation effect is specific to contact with Germany rather than reflecting a generic effect of international exposure (results with French emigration available upon request).

³⁶As mentioned earlier, we lose 15 observations because of the lack of some pages from the ACI yearbook for 1976.

result also holds when Ford and Opel trucks are included in the dependent variable, as shown by the p-values reported at the bottom of the table. The contrast with the car results is sharp: the same provinces, the same elections, the same casualty measures, and the same empirical specification that generate significant negative effects for passenger cars do not lead to any significant result for commercial trucks. This pattern is difficult to reconcile with any supply-side explanation, since disruptions would affect both vehicle types equally. The truck placebo thus points to household consumer taste as the operative channel: wartime violence, mediated through the memory-keeping function of the PCI, shaped what Italian consumers were willing to buy while leaving business purchasing decisions untouched.

Table 7: Placebo: commercial truck registrations

	Y^{foreign} trucks (excl. Ford & Opel)				
	(1)	(2)	(3)	(4)	(5)
PCI share	−0.547 (0.344)	−0.643* (0.353)	−0.614 (0.372)	−0.597 (0.474)	−0.654 (0.431)
PCI share × casualties	−0.000 (0.000)				
PCI share × casualties per capita		0.018 (0.166)			
PCI share × casualties per event			−0.004 (0.023)		
PCI share × casualties per event, 2 nd tercile				−0.163 (0.337)	
PCI share × casualties per event, 3 rd tercile				−0.003 (0.368)	
PCI share × $\ln(1 + \text{casualties})$					0.005 (0.059)
R^2	0.515	0.515	0.515	0.515	0.515
Province FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	894	894	894	894	894
<i>p-value of interaction term, Y^{foreign} trucks (all):</i>					
	0.168	0.836	0.743	0.808 0.669	0.998

Notes: standard errors clustered by province in parentheses with *, **, and *** denoting significance at the 10%, 5%, and 1% level, respectively.

Taken together, the four pieces of evidence make alternative interpretations less plausible. The concentration of the effect on authentically German-branded vehicles and

its attenuation once American-owned Ford and Opel cars are included in the dependent variable rules out a generic aversion to goods manufactured in Germany. The null effect on the overall foreign car share confirms that the mechanism operates within the import basket rather than reducing the propensity to buy foreign goods altogether. The positive interaction between casualties and emigration to Germany shows that direct postwar contact with Germans partially eroded the aversion, a pattern consistent with sentiment but difficult to reconcile with supply-side stories. Finally, the complete absence of any effect for commercial trucks confirms that the mechanism is confined to household consumption goods and leaves business purchasing decisions untouched. The evidence points unambiguously to a precise, persistent, and erodible aversion rooted in wartime memory rather than in any structural feature of the market.

8 Conclusions

This paper studies whether political violence can shape consumer choices long after the violence has ended. We examine postwar Italy, linking cross-provincial variation in civilian casualties caused by Nazi-fascist violence during 1943–1945 to the subsequent purchase of German passenger cars. The evidence indicates that provinces more exposed to wartime violence registered lower shares of German-owned car brands among foreign car registrations when the Italian Communist Party was electorally stronger. The effect is not a general rejection of imports or a standard home-bias mechanism: it operates within the foreign segment of the automobile market and is directed specifically against goods carrying a German national identity.

The broader lesson is that historical shocks persist when they are socially and institutionally maintained. Nazi-fascist violence created the original trauma, but its economic consequences depended on the political infrastructure that kept the memory of that trauma salient. In our setting, the PCI played this role through commemorations, party organizations, veterans’ associations, local newspapers, and a broader anti-fascist public culture. The mechanism we document is therefore not one of individual partisan boycott. It is a mechanism of local public discourse: households made private consumption choices in environments where the meaning of German goods had

been shaped by collective memory.

This interpretation is supported by the structure of the evidence. The estimated effect is concentrated on German-owned passenger-car brands, weakens when American-owned Ford and Opel vehicles manufactured in Germany are included, is not mirrored by a decline in the overall foreign-car share, is attenuated by postwar emigration to Germany, and is absent for commercial trucks. These patterns make it difficult to attribute the results to supply disruptions, generic anti-import sentiment, local economic decline, or business procurement constraints. They point instead to a historically rooted and politically sustained national-origin bias in household consumption.

The paper contributes to the literature on historical persistence by emphasizing the institutional production of memory. Much of the existing work shows that past shocks can have long-run effects on trust, culture, institutions, or political behaviour. Our findings suggest that persistence is not only inherited; it is also maintained. Collective memory is a stock that can depreciate, but its depreciation can be slowed by organizations that reproduce, reinterpret, and transmit it. This distinction helps explain why some historical traumas continue to shape behaviour while others fade. Time alone does not determine the economic afterlife of violence; the survival of memory-keeping institutions matters.

The paper also speaks to the economics of preferences and international exchange. National-origin preferences are often treated as stable background parameters or as slowly moving cultural traits. Our evidence suggests instead that they can be historically produced and politically sustained. The nationality of a good may carry a symbolic cost shaped by past conflict, even after trade relations normalize and markets reopen. This insight is relevant beyond the Italian case. Wars, occupations, colonial violence, diplomatic crises, and state repression may all leave traces in consumer markets when goods retain visible national identities.

Several avenues follow from this argument. Future work could measure memory infrastructures more directly, using data on commemorations, monuments, plaques, school curricula, local newspapers, veterans' associations, or party organizations. It could also study the temporal dynamics of memory, asking whether the disappearance or trans-

formation of memory-keeping institutions (as the PCI was dissolved in 1991) weakens the behavioural effects of historical trauma. Finally, the framework can be extended to other markets in which national identity is visible and economically meaningful, such as tourism, food, cultural products, white goods, or foreign direct investment. However, the social visibility of the good is likely relevant: a car is observed by others in public spaces, making the purchase decision a potential signal of the owner's sentiments, while a refrigerator or washing machine remains largely private.

The central lesson is that the economic legacy of political violence is not written only in destroyed capital, lost income, or demographic loss. It can also survive in the meanings attached to goods. In postwar Italy, Nazi-fascist violence did not prevent German goods from entering the market. It changed which German goods households were willing to buy, especially where political institutions kept wartime memory alive. Ordinary consumption choices can therefore reveal a form of historical persistence that is neither purely material nor purely political, but embedded in the symbolic content of markets.

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Appendix

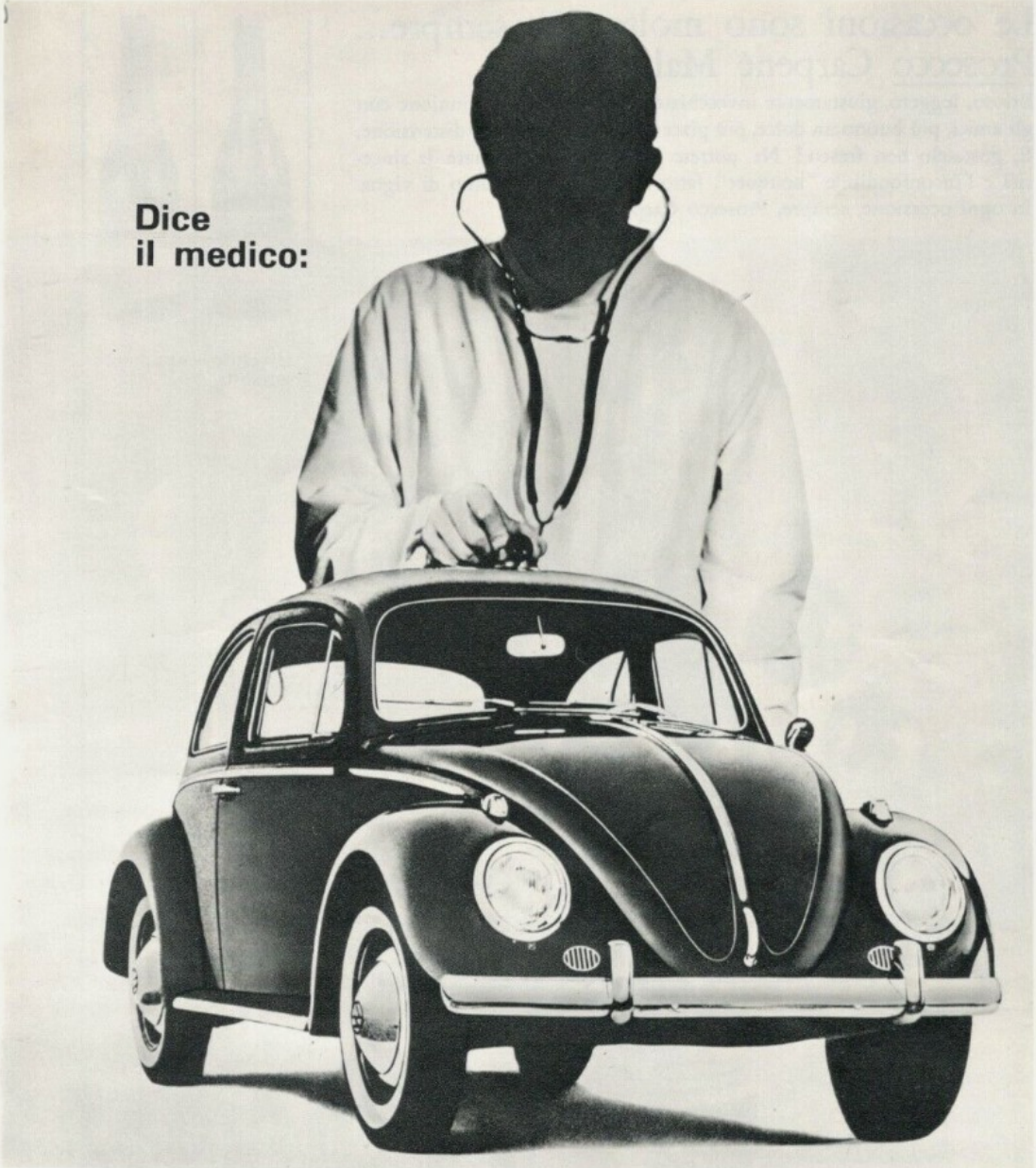
Figure A1: Political posters: PCI and 25 April



Notes: 25 April (*25 Aprile*) is a national holiday in Italy. It commemorates the liberation of Italy from Nazi German occupation and from the collaborationist fascist state of the Italian Social Republic, during the final phase of World War II. The Liberation Day (*Festa della Liberazione*) is distinct from Republic Day (*Festa della Repubblica*), which takes place on 2 June and commemorates the 1946 Italian institutional referendum. The two posters of the Italian Communist Party make explicit reference to the partisan liberation struggle, decades after 1945.

Figure A2: Volkswagen advertising illustration, 1963

**Dice
il medico:**



Berlina 1200 (con paraurti USA) L. 995.000 franco Bologna e Roma I.G.E. compresa. Concessionari in tutte le 92 provincie. Vedere gli indirizzi in tutti gli elenchi telefonici alla lettera "V" - VOLKSWAGEN

"La VOLKSWAGEN è di sana e robusta costituzione"
sicura e, pertanto, raccomandabile anche alle Signore.

VOLKSWAGEN ora di moda anche in Italia

Notes: This 1963 Volkswagen advertising contains subtle messages that echo some of the elements highlighted in the paper: most strikingly, the bottom phrase "VOLKSWAGEN now in fashion also in Italy (*VOLKSWAGEN ora di moda anche in Italia*)" contains a "now" and "also" that implicitly refer to a previous period in which the brand was not fashionable in Italy.

Figure A3: Page from ACI yearbook for 1953

— 10 —

Autovetture « Nuove di fabbrica »

RIEPILOGO

PROVINCIE E COMPARTIMENTI	FABBRICHE COSTRUTTRICI														
	Fiat	Lancia	Alfa Romeo	Bianchi	O.M.	Bandini	Cisitalia	Ermini	Ferrari- Mirco	Ferrari	Giannini	Giur	Innocenti	Isetta	Maserati
Alessandria	1.205	101	48	—	—	—	—	—	—	—	—	—	—	—	6
Asti	246	23	14	—	—	—	—	—	—	—	—	—	—	—	2
Cuneo	810	63	22	—	—	—	—	—	—	—	—	—	—	—	6
Novara	788	139	46	—	—	—	—	—	—	—	—	—	—	—	11
Torino	9.874	791	136	—	—	—	5	—	—	3	—	—	—	2	16
Vercelli	957	143	64	—	—	—	1	—	—	—	—	—	—	1	4
PIEMONTE	13.880	1.260	330	—	—	—	6	—	—	3	—	—	—	3	50
VALLE D'AOSTA	156	35	6	—	—	—	—	—	—	—	—	—	—	—	1
Bergamo	965	88	45	—	—	—	—	—	—	1	—	—	—	—	1
Brescia	2.068	142	67	—	—	—	—	—	—	—	—	—	—	—	1
Como	1.303	145	76	—	—	—	—	—	—	—	—	—	—	1	8
Cremona	646	42	33	—	—	—	—	—	—	—	—	—	—	—	1
Mantova	710	70	22	—	—	—	—	—	—	—	—	—	—	—	—
Milano	14.473	1.032	818	1	12	—	7	—	1	11	—	1	—	9	38
Pavia	1.602	96	53	—	—	—	—	—	—	—	—	—	—	—	—
Sondrio	124	14	3	—	—	—	—	—	—	—	—	—	—	—	—
Varese	1.318	166	101	—	—	—	—	—	—	1	—	—	—	—	5
LOMBARDIA	23.209	1.795	1.218	1	12	—	7	—	1	13	—	1	—	10	54

Notes: The page shows the number of cars “new from production plants” by Italian producers for a sample of 3 regions and their provinces (i.e. there is only one province in Valle d’Aosta). Page 11 contains the remaining producers that do not fit in this page.

Figure A4: Opel advertising in Italy, 1950s–1990s

[illegible]

Panel A. Rekord P1, late 1950s: “la vettura tedesca della General Motors”

LA VETTURA PIÙ CONVENIENTE DEL '63

OPEL KADETT

è una	1000	di cilindrata
ha	4 marce	tutte sincronizzate
scatta fino a	95/h	in terza
tiene i	120/h	su ogni distanza
porta	4/5	persone comode
ha un baule per	6	valigie grandi

...e vi costa meno di un milione "su strada"!

* L. 875.000

liva compresa, in tutta Italia

La Kadett ha il motore anteriore, la trazione posteriore, il raffreddamento ad acqua. È un'automobile tradizionale. Ma se la provate, vi accorgete subito di avere a che fare con una vettura eccezionale, ben vinca. È facile da guidare, ha lo sterzo sciolto e leggero, e vi dà il 95% di visibilità, come una "griglia buca". Il motore è generoso, scattante alla partenza, brillante alla ripresa, silenziosissimo in tutte le marce fino a 4, completamente sincronizzato e con cambio a cliatch.

I freni, straordinari, si danno un gran senso di sicurezza. Comodissima: due ampie portiere, due belle poltrone davanti, un confortevole divanetto.

Il nuovo sistema di sospensione Opel conferisce alla Kadett una eccellente tenuta di strada ed una marcia particolarmente confortevole.

Secondo la migliore tradizione Opel, il baule è grande e profondo. È maggiore nella vettura di questa classe.

Consumo 7 litri ogni 100 Km. non ha mai e poi mai bisogno di ingrassaggio... ed a una Opel!

Con la Kadett avrete sempre un capitale in mano. Tutte le Opel durano di più e per questo mantengono sempre alto il loro valore.

L'AUTOMOBILE FIDATA

prodotta dalla General Motors

NEL VOSTRO INTERESSE PRENOTATEVI SUBITO Concessionari Opel in tutta Italia
General Motors Italia S.p.A. - Milano

© General Motors Corp.

Panel B. Kadett, early 1960s: “*prodotto della General Motors*”

Molta brigata, vita beata nella Kadett

Abitabilità di una 1500. Capacissimo vano porta-bagagli posteriore. Motore da 1078 cc.: a scelta 60 o 55 CV, velocità massima rispettiva 130 km/h e 130 km/h. Fino a 146 km/h col motore potenziato sul Coupé. Freni a disco anteriori e cambio a leva corta centrale a richiesta.

7 modelli: Berlina e Berlina Lusso a 4 o 2 porte, Coupé, Caravan e Caravan Lusso.

Opel **Kadett** GM
la 1000 che va forte

Un prodotto della General Motors

Panel C. Kadett, late 1960s: ice hockey

NUOVA OPEL CORSA

Ci sono persone che hanno una grande passione per le auto.

Ci sono auto che hanno una grande passione per le persone.

Cinture di sicurezza attive

Barre antistriscio

Il side bag

Opel

Chi cerca in un'auto la libertà, la bellezza e l'emozione, trova anche la più grande sicurezza. Trova la nuova Opel Corsa. Un'auto che fa innamorate per la sua personalità unica e che fa di tutto per ricambiare la passione di chi la guida. Perché ha dentro di sé il massimo della tecnologia per la sicurezza ed il comfort: tutto quello che finora trovavi solo su vetture di categoria superiore. Tutto quello che desideri, Corsa può dartelo. • Alzacristalli elettrici • Chiusura centralizzata • Display multifunzionale

- Ventilazione microfilarata e riciclator aia interna • Pediposizione anturto • Cinture con pretensionatore • Barre antistriscio • Servosterzo • Doppio full size airbag • ABS elettronico
- Climatizzatore ecologico • Cambio automatico. Tutti i modelli e motorizzazioni tra cui la Sport con il nuovo e straordinario Ecotec 1.4 16V da 90CV e prezzi speciali. Mercati fino alla consegna, in collaborazione con i Concessionari Opel su tutta la gamma. Chi sceglie una Opel Corsa ha deciso di volersi bene davvero.

Modello	Motori	Potenza (kW)	Velocità massima (km/h)	Consumo (litri/100km) ciclo urbano/extraurbano/misto
Corsa City 3 porte	1.3 81CV	140	182.000	10,5/6,5/7,5
Corsa 5 porte 3 porte	1.3 81CV	140	181.600	10,5/6,5/7,5
Corsa Sport	1.6 100 90CV	140	181.700	10,5/6,5/7,5
Corsa GLS	1.6 100 90CV	140	181.700	10,5/6,5/7,5

Offerta non cumulabile con altre agevolazioni.

Panel D. Corsa, 1994: James Dean edition

Sources: Official Opel archive (<https://opelclassica.it/dagli-inizi-alle-youngtimer/>) for panels A–C; <https://oautoteca.it/tag/james-dean/> for panel D.

Figure A5: Vote shares of Italian Communist Party

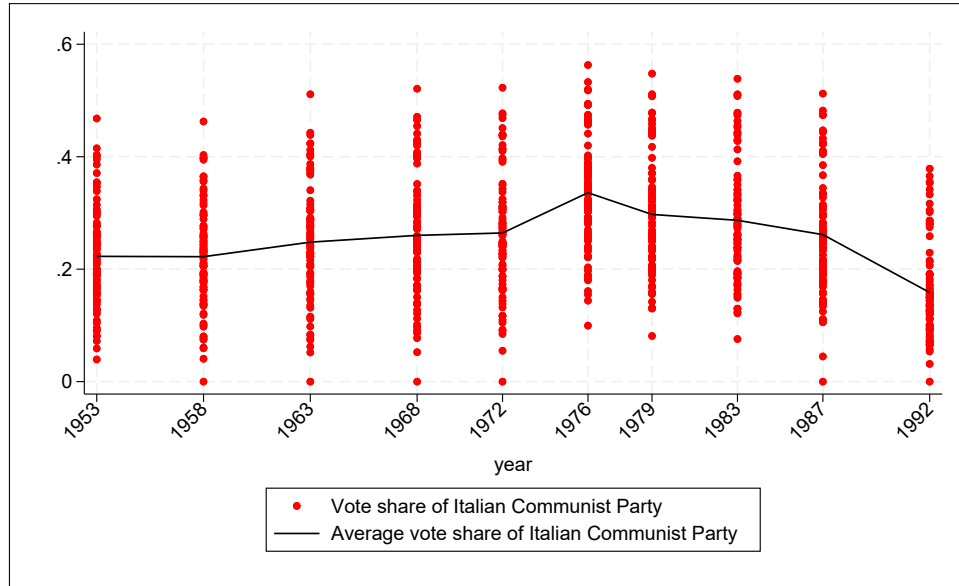


Table A1: Summary statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Population	91	461,624	348,166	140,434	2,175,780
Population density	91	160.89	115.37	31	788
Percentage active population (age 10+)	91	55.00	5.92	42.2	68
Agriculture share	91	53.91	15.87	11.6	77.2
Hunting and fishing share	91	0.49	0.67	0	3.3
Industrial share	91	25.86	10.87	11.6	66.4
Transport and communications share	91	3.46	1.96	1.4	11.8
Commerce share	91	7.27	2.59	3.4	15
Finance and insurance share	91	0.41	0.29	0.1	1.8
Liberal professions and religion share	91	1.24	0.49	0.6	4.1
Public administration share	91	3.99	1.49	2.5	13.4
Private administration share	91	0.37	0.18	0.1	1
Domestic economy share	91	3.00	1.33	1	6.9
Bank deposits	91	429.19	696.47	8.8	5,426
Illiteracy rate (age 6+, 1931)	91	22.41	15.24	1.7	49.8
Jewish deportations per capita	91	0.12	0.23	0	1.55
Emigration to Germany per capita	909	0.88	2.35	0	25.21

Notes: All time-invariant controls refer to the year 1936 except for the illiteracy rate, which is available for 1931; Jewish deportations (per 1,000 people) refer to 1943-45.